

TIP 2021

15. Structure-coherent deep feature learning for robust face alignment.
C. Lin, B. Zhu*, Q. Wang, R. Liao, C. Qian, J. Lu and J. Zhou*

Corresponding-Authored Publications

NeurIPS 2026
Oral (0.4%)

16. Thinking with images as continuous actions: Numerical visual chain-of-thought.
K. Zhao, †B. Zhu, J. Zhou, X. Zhu, Z. Yue and H. Zhang

MM 2026
Oral

17. Layer roles matter: Mitigating modality imbalance in multimodal emotion reasoning via adaptive layer profiling.
P. Huang, P. Song, †B. Zhu, Z. Han, J. Hu and X. Yang

ICML 2026

18. Omni-perception policy optimization for multimodal emotion reasoning.
Z. Han, †B. Zhu, W. Tong, P. Shao, P. Song, X. Wang, J. Chen, L. Lu and X. Yang

CVPR 2026

19. Adapting point cloud analysis via multimodal Bayesian distribution learning.
X. Zhu, L. Yi, S. Wang, W. Zhu, Y. Wu, †B. Zhu and H. Zhang

TIP 2026

20. Hybrid granularity distribution estimation for few-shot learning: statistics transfer from categories and instances.
S. Wang, T. Qi, X. Zhu, Y. Hao, †B. Zhu, H. Zhang and M. Wang

ICLR 2026

21. Real-time motion-controllable autoregressive video diffusion.
K. Zhao, J. Shi, †B. Zhu, J. Zhou, X. Shen, Y. Zhou, Q. Sun and H. Zhang

ICLR 2026

22. Look carefully: adaptive visual reinforcements in multimodal large language models for hallucination mitigation.
X. Zhu, K. Zhao, L. Yi, S. Wang, Z. Wang, †B. Zhu, H. Zhang and X. He

AAAI 2026

23. DEPO: dual-efficiency preference optimization for LLM agents.
S. Chen, M. Zhao, L. Xu, Y. Zhao, †B. Zhu, H. Zhang, S. Zhao and C. Lu

MM 2025
Oral

24. Benchmarking and bridging emotion conflicts for multimodal emotion reasoning.
Z. Han, †B. Zhu, Y. Xu, P. Song and X. Yang

ICCV 2025

25. Unsupervised visual chain-of-thought reasoning via preference optimization.
K. Zhao, †B. Zhu, Q. Sun and H. Zhang

TSG 2019

26. Fault location for radial distribution network via topology and reclosure-generating traveling waves.
S. Shi, †B. Zhu, A. Lei and X. Dong

TSG 2018

27. Fault classification for transmission lines based on group sparse representation.
S. Shi, †B. Zhu, S. Mirsaedi and X. Dong

Others

NeurIPS 2026

28. Beyond SFT-to-RL: Pre-alignment via black-box on-policy distillation for multimodal RL.
S. Wang, W. Huang, X. Yu, Z. Yang, H. Lin, K. Wu, C. Xiao, W. Wang, C. Chen, B. Zhu, Y. Zhang and C. Qin

NeurIPS 2026

29. TopoRefine: Topology-aware correspondence and residual refinement for training-free subject-consistent generation.
Z. Gao, Z. Ti, C. Zhao, B. Zhu, X. Hu, Y. Ge, J. Yang and Y. Tai

TPAMI 2026

30. Generalized kullback-leibler divergence loss.
J. Cui, B. Zhu, Q. Xu, Z. Tian, X. Qi, B. Yu, H. Zhang and R. Hong

TIP 2026

31. Detail++: training-free detail enhancer for text-to-image diffusion models
L. Chen, J. Wang, Z. Pan, B. Zhu, X. Yang and C. Zhang

ACL 2026
Findings

32. AMA: adaptive memory via multi-agent collaboration.
W. Huang, Z. Wang, H. Lin, S. Wang, B. Xu, Q. Li, B. Zhu, L. Yang and C. Qin

ACL 2026

33. Mitigating hallucinations in large vision-language models without performance degradation.
X. Zhu, J. Fang, S. Wang, B. Zhu, Z. Wang, Y. Yang and X. He

ICML 2026

34. Robustifying vision-language models via test-time prompt adaptation.
X. Zhu, H. Wu, S. Wang, B. Zhu, J. Ge, J. Zhang and L. Chen

CVPR 2026

35. Principled steering via null-space projection for jailbreak defense in vision-language models
X. Zhu, B. Zhu, S. Wang, J. Fang, K. Zhao, H. Zhang and X. He

CVPR 2026

36. Few-step diffusion sampling through instance-aware discretizations
L. Yuan, R. Wang, T. Zhao, D. Fu, M. Lei, B. Zhu and C. Zhang

CVPR 2026

37. ACE-merging: data-free model merging with adaptive covariance estimation
B. Xu, H. Wu, H. Lin, W. Huang, B. Zhu, Y. Shu and C. Qin

ICLR 2026

38. CoDi: subject-consistent and pose-diverse text-to-image generation.

	Z. Gao, B. Zhu , L. Yao, J. Yang and Y. Tai	
ICLR 2026	39. GuardAlign: robust safety alignment in multimodal large language models. X. Zhu, B. Zhu , J. Fang, S. Wang, Y. Zhang, X. Wang and X. He	
ICLR 2026	40. Free lunch for stabilizing rectified flow inversion. C. Wang, B. Zhu and C. Zhang	
ICLR 2026	41. Streaming drag-oriented interactive video manipulation: drag anything, anytime! J. Zhou, Y. Zhou, K. Zhao, Q. Xu, B. Zhu , R. Hong and H. Zhang	
AAAI 2026	42. Hierarchical semantic alignment for image clustering. X. Zhu, B. Zhu , Y. Li, J. Fang, S. Wang, K. Zhao and H. Zhang	
Preprint 2025	43. Generative distribution distillation. J. Cui, B. Zhu , Q. Xu, X. Xu, P. Chen, X. Qi, B. Yu, H. Zhang and R. Hong	
NeurIPS 2025 Spotlight (3.2%)	44. Enhancing CLIP robustness via cross-modality alignment. X. Zhu, B. Zhu , S. Wang, K. Zhao and H. Zhang	
ICCV 2025	45. Dynamic multimodal prototype learning in vision-language models. X. Zhu, S. Wang, B. Zhu , M. Li, Y. Li, J. Fang, Z. Wang, D. Wang and H. Zhang	
CVPR 2025	46. Devils in middle layers of large vision-language models: interpreting, detecting and mitigating object hallucinations via attention lens. Z. Jiang, J. Chen, B. Zhu , T. Luo, Y. Shen and X. Yang	
CVPR 2025	47. Stylestudio: text-driven style transfer with selective control of style elements. M. Lei, X. Song, B. Zhu , H. Wang and C. Zhang	
CVPR 2024	48. Classes are not equal: an empirical study on image recognition fairness. J. Cui, B. Zhu , X. Wen, X. Qi, B. Yu and H. Zhang	
AAAI 2023 Oral (10.8%)	49. Leveraging modality-specific representations for audio-visual speech recognition via reinforcement learning. C. Chen, Y. Hu, Q. Zhang, H. Zou, B. Zhu and E. Chng	
Mentoring and Supervision	Xingyu Zhu , Ph.D at USTC <i>Supervisor for papers accepted at MM 2024, NeurIPS 2024/2025, ICCV 2025, AAAI 2026, and ICLR 2026.</i> Kesen Zhao , Ph.D. at NTU <i>Supervisor for paper accepted at ICCV 2025, ICLR 2026 and NeurIPS 2026.</i> Ruoyu Wang , Ph.D. at Westlake University <i>Supervisor for papers accepted at ICCV 2025, NeurIPS 2025 and TPAMI 2026.</i> Zhiyuan Han , Ph.D. at USTC <i>Supervisor for paper accepted at MM 2025 and ICML 2026.</i> Sirui Chen , Ph.D. at Tongji University <i>Supervisor for paper accepted at AAAI 2026.</i> Zhanxin Gao , Ph.D. at Nanjing University <i>Supervisor for paper accepted at ICLR 2026.</i>	
Service	<i>Journal Reviewer:</i> TPAMI, TIP, TMM, TCSVT, TCE, IJCV, Information Fusion <i>Conference Reviewer:</i> CVPR, ICCV, AAAI, MM, UAI, AISTATS, NeurIPS (Top Reviewer 2025), ICLR, ICML (Top Reviewer 2026), ACL.	
Awards	<i>Honorable Mention</i> , Nanyang Speech Forum, Singapore. 2023 <i>AISG PhD Fellowship</i> , National Research Foundation, Singapore. 2021 <i>First Price Scholarship</i> , THU, China. 2018 <i>Scholarship</i> , China Scholarship Council, China. 2014	
Internship	Research Intern, SenseTime, Beijing Mar 2018 – Oct 2018 Research Intern, Tencent YouTu Lab, Shanghai Jul 2018 – Sep 2018 Research Intern, MeiTuan, Beijing Aug 2017 – Dec 2017	

[CV compiled on 2026-09-26]